Chapter 7
Predictability of the May 12, 2008, Wenchuan Earthquake: Insights from the Perspective of 'Dragon King' Theory and 'Nowcasting' Method

Yue Liu, Jiawei Li, Yongxian Zhang, and Zhongliang Wu

Abstract Revisiting the earthquake catalogues, both instrumental and historical, around the central China North–South Seismic Belt, the predictability of the May 12, 2008, Wenchuan $M_S 8.0/M_W 7.8$ earthquake is investigated. This chapter discussed the predictability from the perspective of the 'Dragon King' theory. In terms of the century-scale earthquake catalogue, the Wenchuan earthquake cannot be regarded as a 'Dragon King' event; however, on the decade time scale, it may be regarded as a 'Dragon King' event. In the framework of 'nowcasting earthquakes', hazard of such a devastating earthquake can be described by the 'earthquake potential score' (EPS, up to 94%) and 'potential magnitude' (M_P, up to 7.4) just prior to the occurrence of this event.

Keywords The 2008 Wenchuan earthquake · 'Dragon King' theory · Nowcasting earthquakes · Seismic hazard · Earthquake potential score (EPS) · North–South Seismic Belt

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Y. Liu · Y. Zhang (✉) · Z. Wu (✉)
Institute of Earthquake Forecasting, China Earthquake Administration, No. 63 Fuxing Avenue, Beijing, China
e-mail: yxzhseis@sina.com

Z. Wu
e-mail: wuzl@cea-igp.ac.cn

J. Li · Z. Wu
Institute of Risk Analysis, Prediction and Management (Risks-X), Academy for Advanced Interdisciplinary Studies, Southern University of Science and Technology, No. 1088 Xueyuan Avenue, Shenzhen, China
e-mail: lijw@cea-igp.ac.cn

Y.-G. Li et al. (eds.), *Scientific Investigation of Continental Earthquakes and Relevant Studies*, https://doi.org/10.1007/978-981-96-2826-1_7

7.1 Introduction

Since the occurrence of the May 12, 2008, Wenchuan M_S8.0/M_W7.8 earthquake (Chen & Booth, 2011; Wu & Ma, 2014), vast studies have been carried out on its tectonics, preparation, source process, disasters, aftershock sequence, and their geophysical and geological implications (e.g., Lay, 2019; Xu & Li, 2019). For such a devastating earthquake, one of the outstanding questions is whether it could be regarded as an unexpected extreme event, which needs the analysis of observational data from different perspectives. In this chapter we revisit this question in the views of 'dragon king' theory (Eliazar, 2017; Glette-Iversen & Aven, 2021; Lin et al., 2018; Pisarenko & Sornette, 2012; Premraj et al., 2021; Sachs et al., 2012; Sornette & Ouillon, 2012; Yukalov & Sornette, 2012) and 'nowcasting earthquakes' (Luginbuhl et al., 2018; Rundle et al., 2016, 2018, 2020, 2021; Varotsos et al., 2005, 2011), two approaches to seismic hazard assessment and/or earthquake forecast proposed since the recent two decades, originated from the physics of complexity and applied to earthquake science.

7.2 Earthquake Catalogues Used for the Analysis

For the Wenchuan earthquake, with its process of seismogenesis and earthquake preparation covering a long-time duration with at least a century scale, and a large spatial range related to the central China North–South Seismic Belt, one of the critical datasets for the analysis of its predictability is the earthquake catalogue, covering both instrumental and historical records. This study uses the Collection of Earthquake Catalogues of China $M \geq 5.0$ (January 1, 1900 to December 31, 2020), authorized by the Department for Earthquake Monitoring and Forecasting, China Earthquake Administration (CEA) in 2023[1], as a continuation of a long-term endeavor to have a re-calibrated and unified earthquake catalogue for analysis (Earthquake Disaster Prevention Department, State Seismological Bureau, 1995; Department of Earthquake Disaster Prevention, China Seismological Bureau, 1999). We used the recent century of the catalogue, from January 1, 1920, to December 31, 2019.

Figure 7.1 shows the location of the Wenchuan earthquake and the distribution of earthquakes with $M \geq 6.0$ for the whole region in the recent century from the above-mentioned catalogue, together with the aftershocks of the Wenchuan earthquake (from the online catalogue of the China Earthquake Networks Center (CENC), delineating the rupture zone). The whole region basically covers the central China North–South Seismic Belt which is located in the eastern part of the Qinghai-Tibet Plateau (the Tibetan Plateau). The region, extending approximately 21° N to 40° N and 96° E to 110° E, has been experiencing strong seismic activity since records available. A total of 188 $M \geq 6.0$ earthquakes have occurred in the region during

[1] Department for Earthquake Monitoring and Forecasting, China Earthquake Administration (CEA), 2023, Open file no.12.

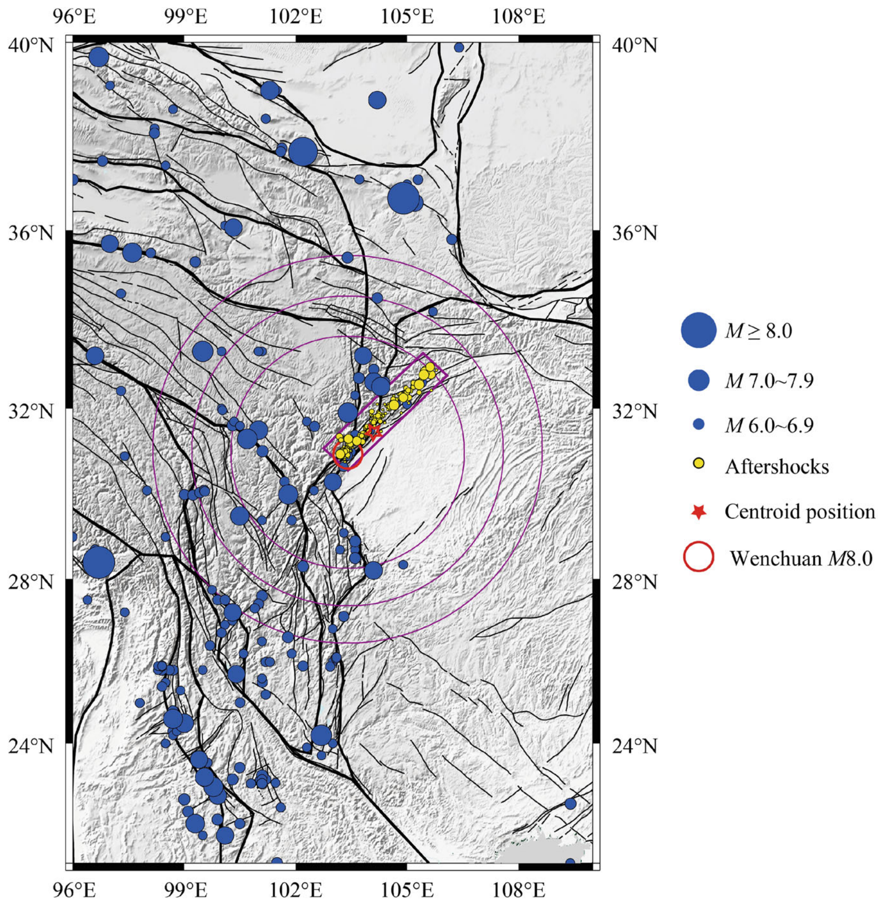

Fig. 7.1 Distribution of earthquakes ($M_S \geq 6.0$, from 01/01/1920 to 12/31/2019) within the central China North–South Seismic Belt. Blue dots represent the epicenters. The red circle shows the epicenter of the 2008 Wenchuan $M_S 8.0$ earthquake, with red star showing its centroid. The three purple circles centering at the epicenter of the Wenchuan earthquake are the regions for the nowcasting analysis, with radius 300 km, 400 km and 500 km, respectively. Rectangle in purple plots approximately the zone of aftershocks, with the yellow dots showing the aftershocks with $M_L \geq 4.5$ in which M_L is the Chinese local magnitude

the recent century (01/01/1920–12/31/2019), with four $M \geq 8.0$ and thirty $M7.0$–7.9 events. Earthquakes of $M \geq 7.5$ are listed in Table 7.1. Magnitude has been defined and measured in an organized and authorized way in China, as represented by the national standard in 1999 and 2017[2], respectively. In the first catalogue, earthquakes are quantified with M_S, the Chinese version of surface wave magnitude (Bormann et al., 2007).

[2] *General Ruler for Earthquake Magnitude*. GB 17740-2017, update of GB 17740–1999.

Table 7.1 Earthquakes of $M \geq 7.5$ occurred in the central China North–South Seismic Belt since 1920 (data is from the Collection of Earthquake Catalogues of China $M \geq 5.0$ (January 1, 1900 to December 31, 2020))

	Date Beijing Time (UTC + 8)	Latitude (°N)	Longitude (°E)	Magnitude	Location
1	1920-12-16	36.7	104.9	8.5	Haiyuan, Ningxia
2	1927-05-23	37.7	102.2	8.0	Gulang, Gansu
3	1932-12-25	39.7	96.7	7.6	Yumen, Gansu
4	1933-08-25	31.9	103.4	7.5	Maoxian, Sichuan
5	1937-01-07	35.5	97.6	7.5	Maduo, Qinghai
6	1947-03-17	33.3	99.5	7.7	Dari, Qinghai
7	1950-08-15	28.4	96.7	8.6	Zayu, Xizang
8	1955-04-14	30.0	101.8	7.5	Garzê, Sichuan
9	1970-01-05	24.2	102.7	7.8	Tonghai, Yunnan
10	1973-02-06	31.3	100.7	7.6	Luhuo, Sichuan
11	2008-05-12	31.0	103.4	8.0	Wenchuan, Sichuan

7.3 Predictability of the Wenchuan Earthquake in the Perspective of 'Dragon King' Theory

'Dragon King' theory (Eliazar, 2017; Glette-Iversen & Aven, 2021; Lin et al., 2018; Pisarenko & Sornette, 2012; Premraj et al., 2021; Sachs et al., 2012; Sornette & Ouillon, 2012; Yukalov & Sornette, 2012) deals with the extreme events in a variety of natural and/or social systems. As a statistical outlier, a 'dragon king' event deviates significantly from the power law distribution and exhibits some predictability.

The Zipf distribution which has been widely used in various fields (Li, 2002; Saichev et al., 2010) is one of the efficient tools for the identification of 'dragon king' events (Pisarenko & Sornette, 2012). Operationally, the data subject to analysis is ordered from large to small values, and the rank versus the quantity is visualized in a double log plot. For earthquakes, due to the definition of magnitudes which are the logarithm, or the order of magnitudes, of seismic moment or radiated energy, the rank ordering plot naturally becomes single log. It has been proved (Sornette et al., 1996) that if a power law distribution exists, the rank-ordering plot will exhibit a linear distribution on the double log plot. For the cases of only a few data samples, rank-ordering analysis may avoid the imbalance between the numerous data points of small values and the few data points of large values, which shares similarity with the most-likelihood estimation of the Gutenberg-Richter b-value (Aki, 1965).

For the data set with N elements, $\{x_i, i = 1, 2, \ldots N\}$, when each sub-set of data is generated by excluding the first n data points, in which n changes from 0 to a number much smaller than N, the data from the $(n + 1)$th to the Nth point fit a power law distribution, and the first n points may be or may not be the 'dragon king' events. The

b-value for the $(n + 1)$th to the Nth points can be estimated using the most-likelihood estimator, represented by the mean magnitude (Aki, 1965) as

$$b = \frac{\log_{10} e}{\overline{M} - M_{\mathrm{C}}} \tag{7.1}$$

in which M_{C} is the cutoff magnitude, $\overline{M}$ is the average magnitude. Null hypothesis is that the first n points, although somehow deviating from the extension of the power law scaling, are also the outputs from the power law distribution, albeit with some fluctuations. Rejection of the null hypothesis can be decided either by visual inspection or by quantitative formulation (Pisarenko & Sornette, 2012).

Figure 7.2 shows the rank-ordering plot for earthquakes no less than $M6$, with the Wenchuan earthquake marked as a star. It may be seen that, for the whole North–South Seismic Belt and for the recent century, the Wenchuan earthquake ranks the third among all the earthquakes. For smaller spatial ranges, the Wenchuan earthquake, although ranks the first, still falls into the power law scaling. Visual inspection can easily exclude the possibility that the Wenchuan earthquake is a 'Dragon King' event. Rather it fits in the power law scaling, with the b value 0.86 ± 0.06. This implies that, in a long-term sense with century scale, intrinsic unpredictability exhibits associated with this event, as indicated by the model of self-organized criticality (SOC, Bak & Tang, 1989; Ito & Matsuzaki, 1990; Sornette & Sornette, 1989).

The scaling law of earthquakes depends on the samples subject to consideration. Similar to the formulation of Molchan and Kronrod (2005) and Nekrasova et al. (2011), that

$$\log N = a - bM + c \log L \tag{7.2}$$

in which L is the scale of spatial range and c is a constant, we may have

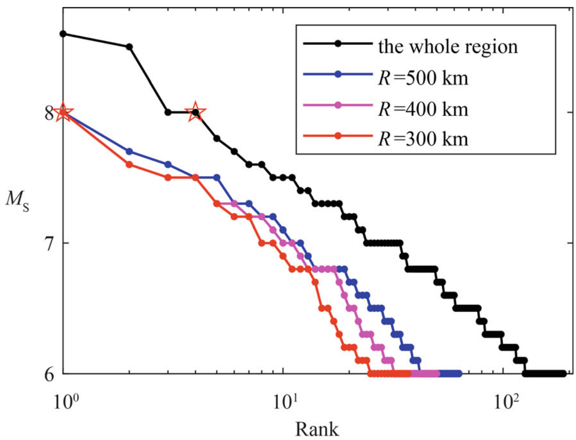

Fig. 7.2 Rank-ordering plot for $M_{\mathrm{S}} \geq 6.0$ earthquakes. Black line represents the result for the north–south seismic belt. Red, pink and blue lines show the results within the circular areas centered at the epicenter of the Wenchuan earthquake with radius 300 km, 400 km, and 500 km, respectively. Red stars represent the 2008 Wenchuan earthquake

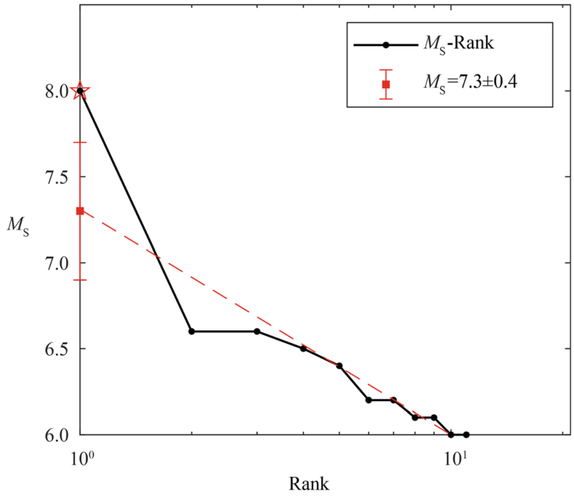

Fig. 7.3 Rank-ordering plot for $M_S \geq 6.0$ earthquakes during the decade before and including the Wenchuan earthquake (05/12/1998–05/12/2008). The red star represents the 2008 Wenchuan $M_S 8.0$ mainshock. The red dashed line fits the power-law scaling

$$\log N = a - bM + d \log T \tag{7.3}$$

in which T is the time duration subject to analysis and d is a constant. Equations (7.2) and (7.3) reflect the self-affine character of seismicity. Figure 7.3 shows the result for the time duration from 05/12/1998 to 05/12/2008. It can be seen that within this decade, the Wenchuan earthquake is likely a 'dragon king' event. The expected maximum magnitude from the extension of the power law scaling is 7.3 ± 0.4. Compared to magnitude 8.0 of the Wenchuan earthquake, the null hypothesis that the Wenchuan earthquake is also an output of the same power law scaling can be rejected with a confidence level higher than 95%. This implies that, for a decade scale, the Wenchuan earthquake seemed possessing some predictability. As a matter of fact, precursor-like anomalies were observed, albeit somehow regrettably with majority of them based on retrospective case studies, with a characteristic time scale ranging from one year to a decade (Ma & Wu, 2012).

7.4 Hazard of the Wenchuan Earthquake in the Perspective of 'Nowcasting'

Although the Wenchuan earthquake was not predicted, and the analysis in Sect. 7.3 indicates some of its intrinsic unpredictability (in the sense of 'Dragon King' theory), whether the hazard of this devastating event could be assessed is one of the scientific problems with critical importance for seismic disaster resilience. And the 'nowcasting' approach (Luginbuhl et al., 2018; Rundle et al., 2016, 2018, 2020, 2021) may be of help in this respect.

In the 'nowcasting' approach (Rundle et al., 2016, 2018, 2020, 2021), the magnitude of the target events, M_λ, and the magnitude of small earthquakes, M_σ, which are all larger than the completeness magnitude, have clear relations with the number of events, described by the well-known Gutenberg-Richter distribution. If N_σ is the average number of small earthquakes greater than M_σ, and N_λ the average number of the large (target) earthquakes greater than M_λ, then

$$N_\sigma = 10^a 10^{-bM_\sigma} \tag{7.4}$$

$$N_\lambda = 10^a 10^{-bM_\lambda} \tag{7.5}$$

The average number of small earthquakes N between large earthquakes is thus

$$N = \frac{N_\sigma - N_\lambda}{N_\lambda} = 10^{b(M_\lambda - M_\sigma)} - 1 \tag{7.6}$$

According to the 'nowcasting' approach (Rundle et al., 2016, 2018, 2020, 2021), the potential for large earthquakes (the target events) is estimated by the cumulative distribution function (CDF) of the small events with magnitude between M_σ and M_λ. The time-dependent value of CDF, $p[n \leq n(t)]$, using the current count of small earthquakes $n(t)$ at time t, can then be obtained, in which t is the calendar time since the last large earthquake.

The number counting of small events leads to another kind of 'time', the 'natural time' (Varotsos et al., 2005, 2011), for the analysis of earthquake recurrence. The 'earthquake potential score' (*EPS*) at time t is defined by (Rundle et al., 2016, 2018, 2020, 2021):

$$EPS = p[n \leq n(t)] \tag{7.7}$$

which is between 0 and 1, increases with time since the last target earthquake, and resets to zero when the next large earthquake occurs.

Figure 7.4 plots the EPS on 2008/05/11, with a value of 80%. In the calculation, following previous works (e.g., Rundel et al., 2021), the target region is selected as a circular area, with a radius of 500 km centered at the epicenter of the mainshock. To investigate the influence of parameter selection, sensitivity tests have been conducted by changing the value of M_λ (7.0, 7.1, … 7.5, respectively), the radius of the target region (300, 400, 500 km, respectively), and the position of the center (epicenter or centroid), for the circular target area. Table 7.2 lists the EPS on 2008/05/11 for different parameter settings. It turns out that the EPS ranges from 20% to 94%. The *EPS* in a larger area is higher than that in a smaller area, which might be related to the characteristics of the seismogenic process of this earthquake, that moment deficits (rather than only seismic activity) play an important role (Wang et al., 2011).

Following Rundle et al. (2020), the 'potential magnitude' M_{P} for the forthcoming earthquake can be estimated by

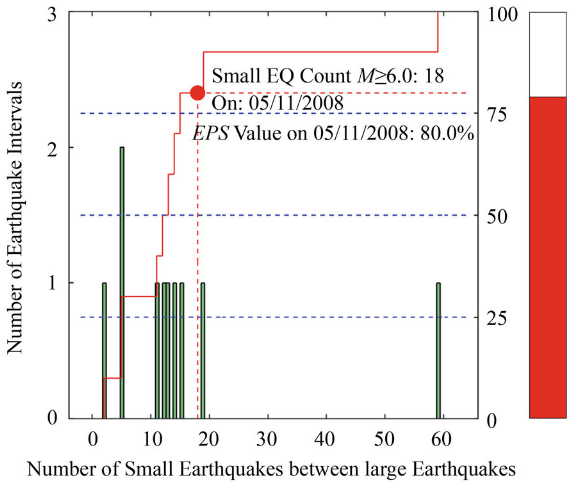

Fig. 7.4 Earthquake potential score (EPS) just before the 2008 Wenchuan earthquake within 500 km centered at its epicenter. Figure captions are similar to that of Rundle et al. (2021). The result is for $M_\lambda \geq 7.5$ earthquakes within Wenchuan source region after $M7.6$ on 1973/2/6. The green bars are the histograms of the number of small earthquakes ($6 \leq M_\sigma \leq 7.4$) between the large earthquakes. The red curves are the CDF corresponding to the histogram. The red dot corresponds to the number of small earthquakes since the last large earthquake. 'Thermometer' on the right is a visual representation of the EPS

Table 7.2 Sensitivity test of EPS (%) just before the 2008 Wenchuan earthquake

		EPS (%) on 2008/05/11 and M_P when M_λ is different					
		$M_\lambda = 7.5$	$M_\lambda = 7.4$	$M_\lambda = 7.3$	$M_\lambda = 7.2$	$M_\lambda = 7.1$	$M_\lambda = 7.0$
(1)	$R = 500$ km	80.0/7.4	75.0/7.4	94.1/7.4	60.0/7.2	63.6/7.2	84.9/7.2
	$R = 400$ km	60.0/7.3	58.3/7.3	70.6/7.3	55.0/7.0	54.6/7.0	66.7/7.0
	$R = 300$ km	30.0/7.1	33.3/7.1	52.9/7.1	25.0/6.5	27.3/6.5	45.5/6.5
(2)	$R = 500$ km	70.0/7.4	66.7/7.4	70.6/7.4	55.0/7.0	54.6/7.0	78.8/7.0
	$R = 400$ km	30.0/7.2	33.3/7.2	52.9/7.2	30.0/6.7	36.4/6.7	54.6/6.7
	$R = 300$ km	30.0/7.1	33.3/7.1	52.9/7.1	20.0/6.3	22.7/6.3	30.3/6.3

Note (1) Results for the target region being a circle with the epicenter as its center; (2) Results for the target region being a circle with the centroid as its center

$$M_P = M_\sigma + (1/b)\log_{10} N_P \tag{7.8}$$

where N_P is the number of earthquakes with magnitude range $[M_\sigma, M_\lambda)$ since the last large earthquake with magnitude greater than or equal to M_λ. The potential magnitude M_P for the study region before the occurrence of the Wenchuan earthquake ranges from 6.3 to 7.4. Similarly, the M_P for a larger area is higher than that for a smaller area.

Note that limits of the number of samples prevent from selecting the target magnitude greater than 7.5, which is a problem in this analysis. In this sense, this estimate provides the assessment of seismic hazard with a 'bottom-line' result. But the EPS reaches 94% and the potential magnitude reaches 7.4 means that in the perspective of 'nowcasting earthquakes', the hazards of the Wenchuan earthquake are significant.

7.5 Conclusions and Discussion

Focusing on the Wenchuan earthquake, we inspected the frequency-magnitude scaling of earthquakes to explore whether this great earthquake could be regarded as a 'Dragon King' event. We used the rank-ordering representation to examine the scaling. From the data it seems that on a century scale, the Wenchuan earthquake fits the power law scaling, and cannot be regarded as a 'Dragon King' event. This implies that, in a long-term sense, this great earthquake has an intrinsic physical unpredictability as revealed by, e.g., the self-organized criticality (SOC) model. For a decade scale, the Wenchuan earthquake deviates from the power law scaling as an outlier, and can be regarded as a 'Dragon King' event, therefore has some predictability.

Gutenberg-Richter power law plays a crucial role in the analysis, with the 'Dragon King' theory stressing the deviation, and the 'nowcasting' approach stressing the power law itself. For this great earthquake, its hazards can be characterized by the 'earthquake potential score' (EPS) and the 'potential magnitude' M_{P} just prior to this earthquake. The results indicate that, even if only an $M7.6$ event is selected as the last event in the earthquake cycle, the EPS before the day when this great earthquake stroke had reached 94% and the M_{P} had reached 7.4. The results, although regrettably of minor help for the preparedness of the Wenchuan earthquake, might help in the future in coping with similar great earthquakes.

It may not be so appropriate for this chapter to simply apply the 'Dragon King' theory and the 'nowcasting' method, to discuss the limits of these two present approaches. On the other hand, however, discussing the future research agenda related to these two approaches may help in deepening the understanding of earthquake predictability. For the seismic dragon king theory, the dependence on the selection of the time duration and spatial range has to be accounted for, and it is somehow questionable whether magnitude is a good index in the identification of dragon king event/s. Compared to other fields such as economics which have successful cases of predicting the catastrophic events (Sornette & Ouillon, 2012), it may make sense to ask the question whether, financial fluctuation has intrinsic differences from an economic crisis, what is the intrinsic difference between a large earthquake and small ones. For the nowcasting approach, although enhanced nowcasting (Rundle et al., 2020) is a solution to the problem of the completeness of earthquake catalogues, for the long-duration earthquake catalogues which mix historical and instrumental ones, this is still a problem in need of sophisticated statistical treatment. The role of aftershocks may be another issue in need of consideration. Nevertheless, the two approaches, originating from the physics of complexity, may provide the discussion on the predictability of earthquakes with new insights, as shown in the case of the Wenchuan earthquake.

Acknowledgements This work is supported by the National Natural Science Foundation of China (grant number U2039207). Thanks are due to Prof. J. B. Rundle (University of California, Davis) and Prof. Didier Sornette (Southern University of Science and Technology, Shenzhen) for stimulating discussion and advice on the related theories and algorithms.

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